VAR-DT8MCustomBoard

Variscite

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12.	PINMUX J1 & J2 & J3
13.	CAN FD INTERFACE

Disclaimer:

SchematicS are for reference only.
Variscite LTD provides no warranty for the use of these schematics.
Schematics are subject to change without notice.

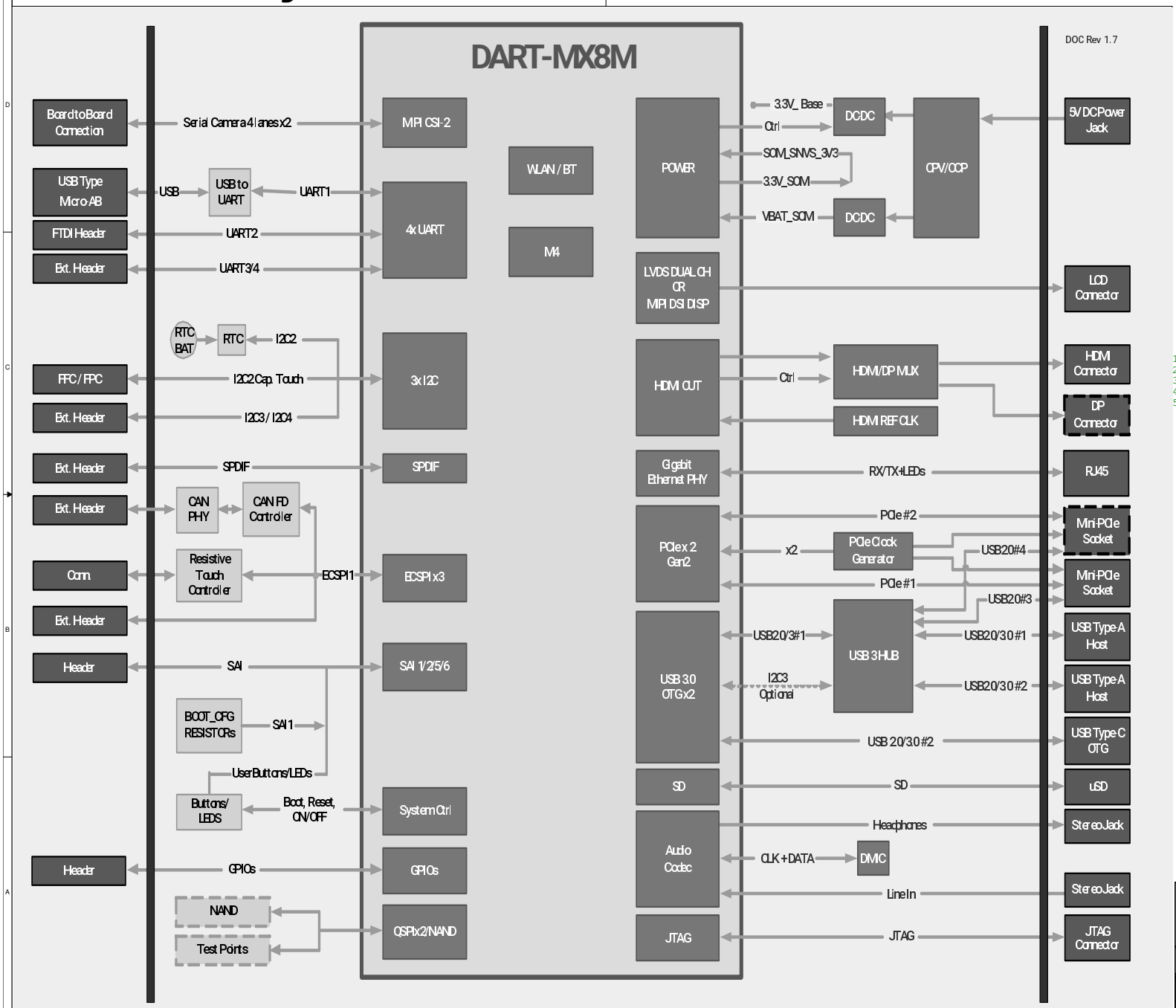
Revision History

Document	Carrier	Description
1.0	1.0	INITIAL
1.1	1.1	1st Release
1.2	1.2	Schottky_SSMINI - Replace symbol (swapped pin 1 and 2 to match silk) e.g. D1 DART_J1.31 - Update connector for NVCC_ENET pin R110 - Assemble for PMIC_ON_REQ to go low for >130ms BASE_PER_3V8 - feedback taken from SOM_VBAT (close to SOM) -Rev1.1. add on wire R159 and R156 - Remove to allow FPF2193 auto restart R176 - Replaced to 17.8K to allow for 5.4V power supply C157 - Added on input power eFUSE - filter glitches R65 - Remove - Part of boot config - not required. Open Solder mask and add thermal pad under SOM
1.3	1.3	Added support for Basler MIPI-CSI camera DP - Align with NXP reference design DART-MX8M-MINI notes/block diagram and symbol added. Added CAN-FD to SPI bridge circuitry
1.4	1.3A	Limit DMIC_DATA to 1.8V swing using a voltage divider Overdriving DMIC_DATA (>1.8V) (applicable only when recording DMIC input) will generate noise on Headphone output.
1.5	1.3B	Added DisplayPort connector J20 and remove disclaimer note
1.6	1.4	Fix Layout for DMIC_DATA voltage divider Add page 13. to Content list
1.7	1.4	Correct DART-MX8M and DART-MX8M-MINI J2 symbols for pin names on J2.2 and J2.14; See Pinmux changes for HPLOUT & DMIC_CLK nets.

Variscite

Title: 01. Cover			
Size: A3	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Monday, August 26, 2019		Sheet 1 of 15	

02A. Block Diagram – DART-MX8M



I2C BUS ADDRESS:

I2C1: Internal to SOM

I2C2: PU - 10K on U8
10K on custom
0x54 BOARD ID EEPROM Page0
0x55 BOARD ID EEPROM Page1
0x68 RTC
0x38 CAPACITIVE TOUCH CTRLR
0x3D USB-C CC Logic PTN5150AHXMP
0x3C CSI P1 Camera (1V8) OV5640

I2C3: PU - 5K on SOM
0x60 SOM - Int. power ctrl.
0x2D USB3 HUB
0xXX Header J12

I2C4: PU - 10K on U8
10K on custom
0x3C CSI P2 Camera (1V8) OV5640
0xXX Header J12
0xXX mPCIe J23 & J32

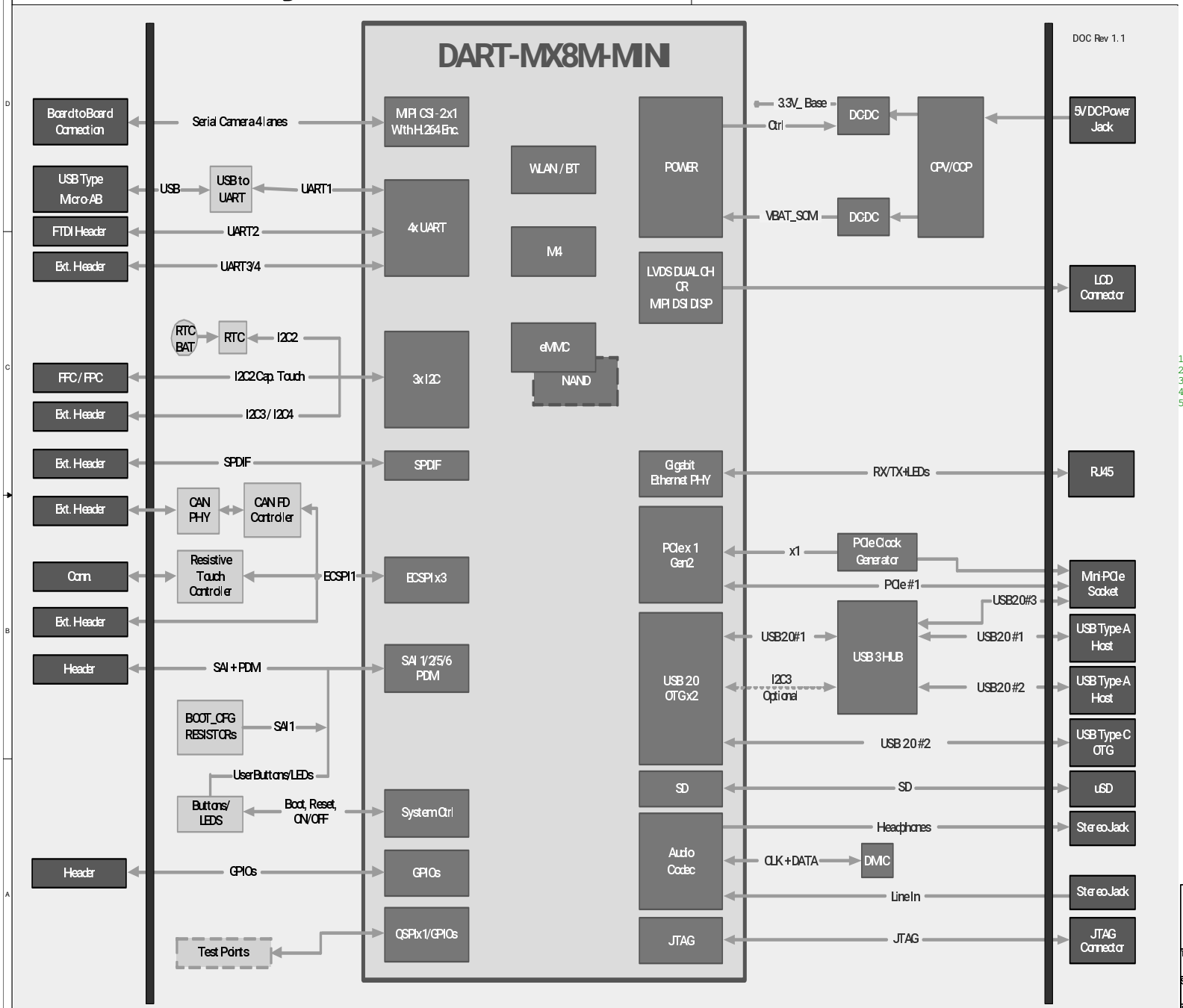
Important Notes:

1. Length match for HS signals according to SOM DS
2. USB routed as 90 ohm Diff pairs
3. PCIe/SATA routed as 85 ohm Diff pairs
4. LVDS routed as 100 ohm Diff pairs
5. Other fast changing signals routed as 50 ohm

Variscite

Title: 02A. Block Diagram with DART-MX8M			
Size: A3	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Thursday, August 08, 2019		Sheet 2 of 15	

02B. Block Diagram – DART-MX8M-MINI



I2C BUS ADDRESS:

I2C1: Internal to SOM

I2C2: PU - 10K on U8
10K on custom
0x54 BOARD ID EEPROM Page0
0x55 BOARD ID EEPROM Page1
0x68 RTC
0x38 CAPACITIVE TOUCH CTRLR
0x3D USB-C CC Logic PTN5150AHXMP
0x3C CSI P1 Camera (1V8) OV5640

I2C3: PU - 5K on SOM
0x1A SOM - Int. CODEC
0x2D USB3 HUB
0xXX Header J12

I2C4: PU - 10K on U8
10K on custom
0x3C CSI P1 Camera (1V8) OV5640
0xXX Header J12
0xXX mPCIe J23 & J32

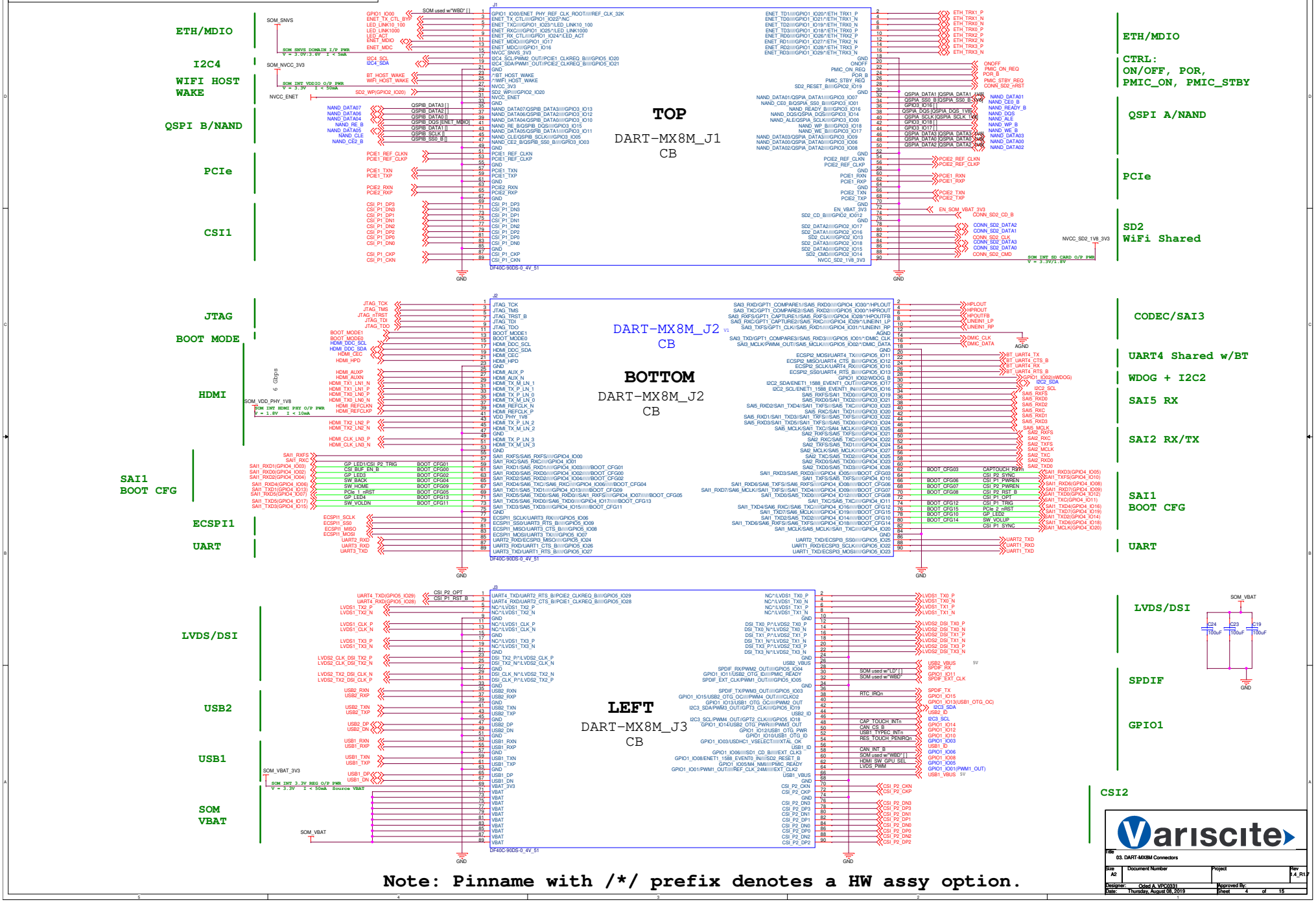
Important Notes:

1. Length match for HS signals according to SOM DS
2. USB routed as 90 ohm Diff pairs
3. PCIe/SATA routed as 85 ohm Diff pairs
4. LVDS routed as 100 ohm Diff pairs
5. Other fast changing signals routed as 50 ohm

Variscite

Title: 02B. Block Diagram with DART-MX8M-MINI			
Size: A3	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Monday, August 26, 2019		Sheet 3 of 15	

03A – DART-MX8M Connectors

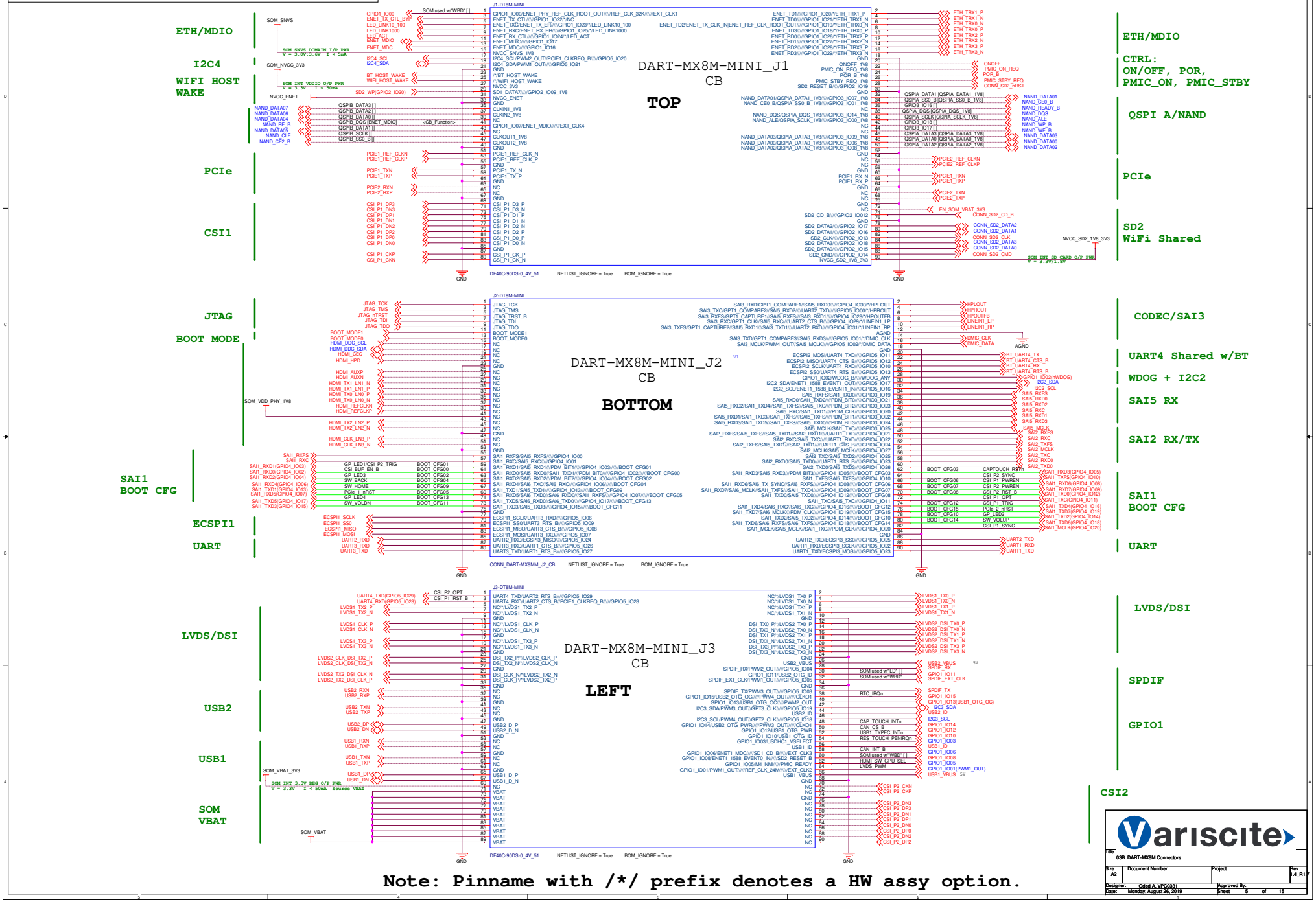


Variscite

Title: 03. DART-MX8M Connectors			
Size: A2	Document Number	Project	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Thursday, August 08, 2019		Sheet 4 of 15	

03B – DART-MX8M-MINI Connectors

*** Dotted nets – Functionality differ from DART-MX8M. ***

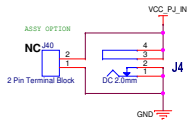


Note: Pinname with /*/ prefix denotes a HW assy option.

Title	03B. DART-MX8M Connectors
Size	A2
Document Number	
Project	
Rev	1.4_R1.7
Designer:	Oded A. VPC0331
Approved By:	
Date:	Monday, August 26, 2019
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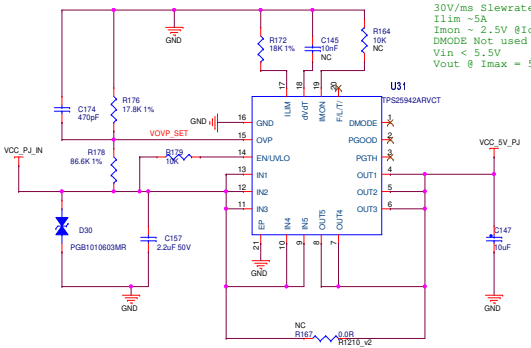
04. Power, RTC,Board ID

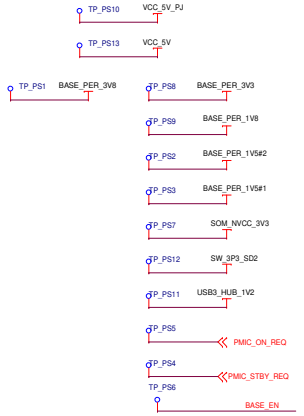
5VDC/4A POWER JACK



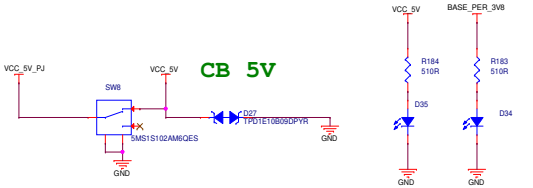
PWR JACK 5V IN OVP/OCP

30V/ms Slewrate based on int. circuitry
Ilim ~5A
Imon ~ 2.5V @Iout=5A
DMODE Not used
Vin < 5.5V
Vout @ Imax = 5.2V

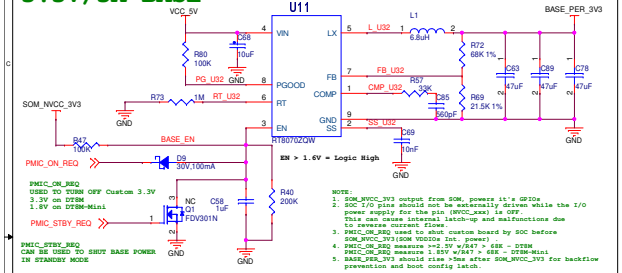




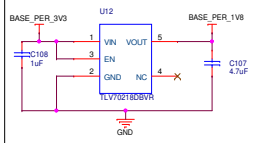
POWER SW/(ON/OFF)



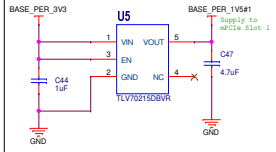
3.3V/3A BASE



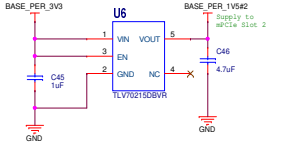
1.8V/0.3A BASE



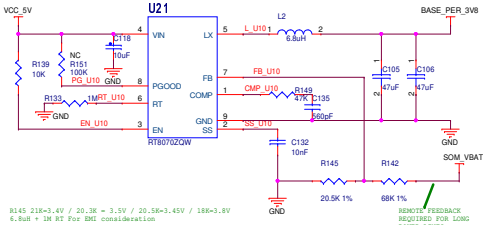
1.5V/0.3A #1 BASE



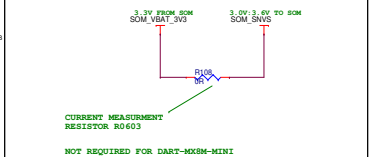
1.5V/0.3A #2 BASE



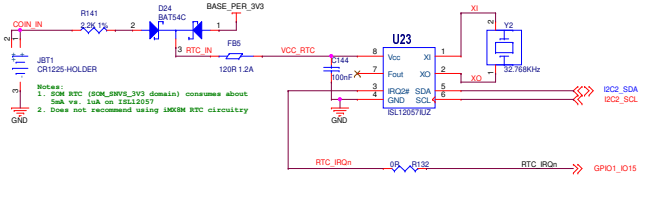
3.45V/3A FROM PWR JACK



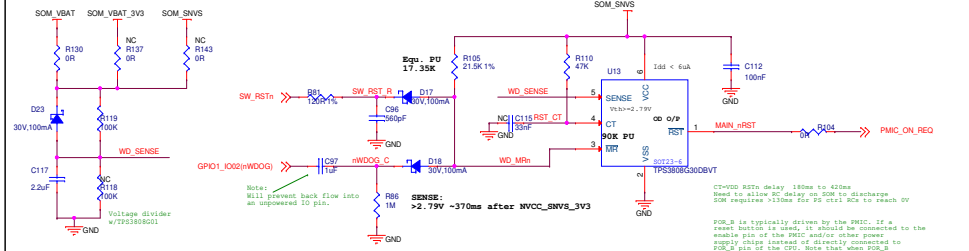
SOM SNVS



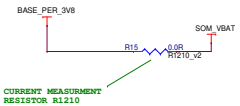
RTC BATTERY



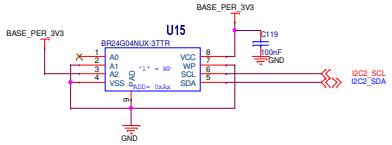
RESET & WATCHDOG



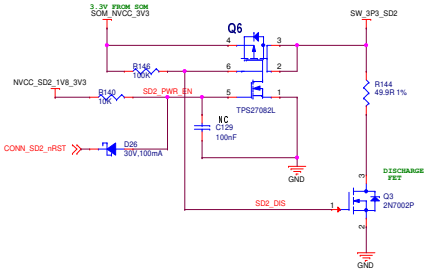
SOM PWR



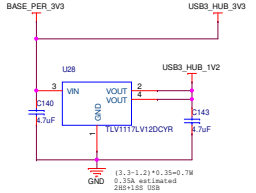
BOARD ID



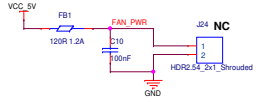
SD POWER



USB3 HUB POWER



FAN : 5V/0.2A



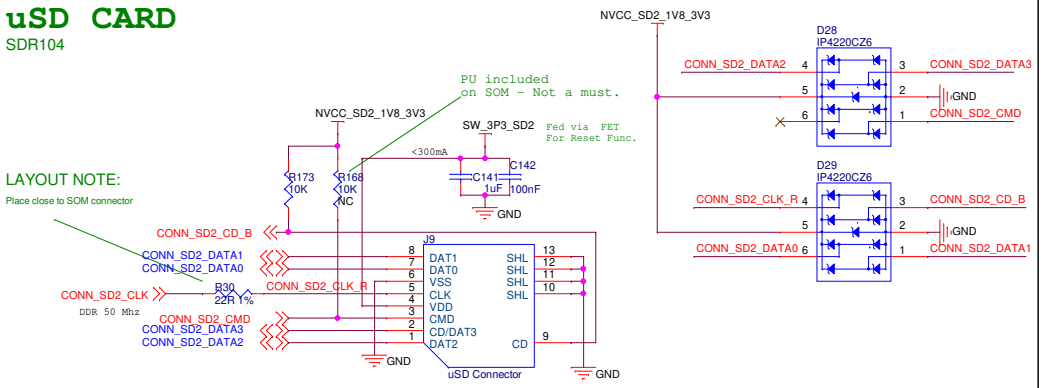
Variscite

Title: 04. Power, RTC,Board ID			
Size: A2	Document Number	Project	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Thursday, August 08, 2019		Sheet 6 of 15	

05. ETH, uSD, AUDIO, MIPI-CSI

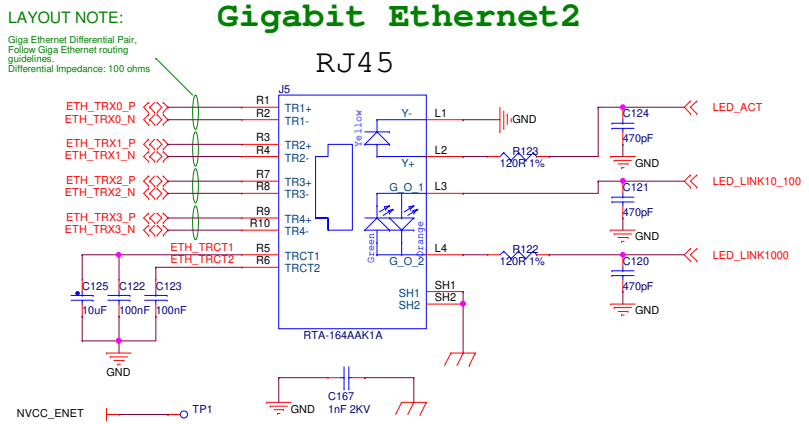
uSD CARD

SDR104

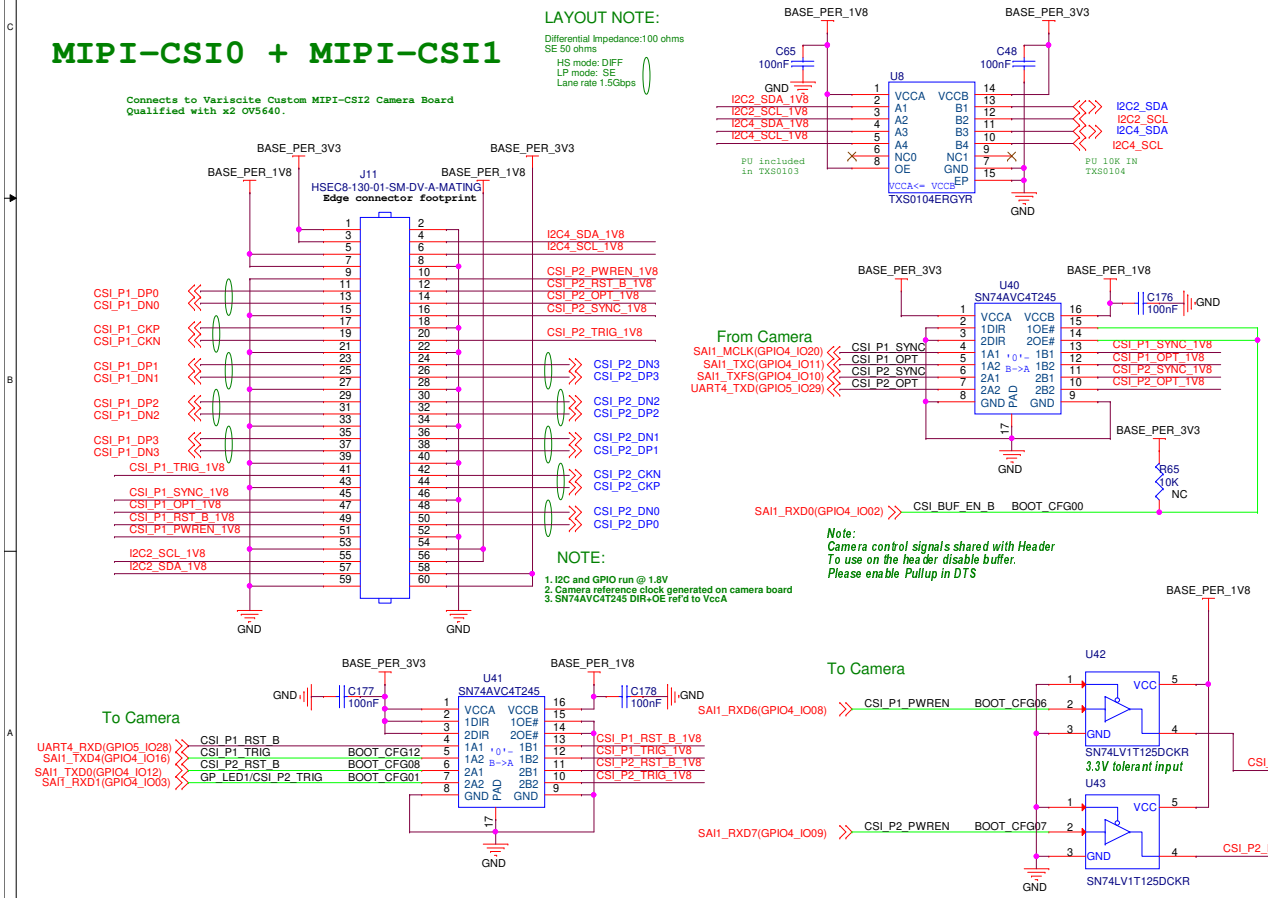


Gigabit Ethernet2

RJ45

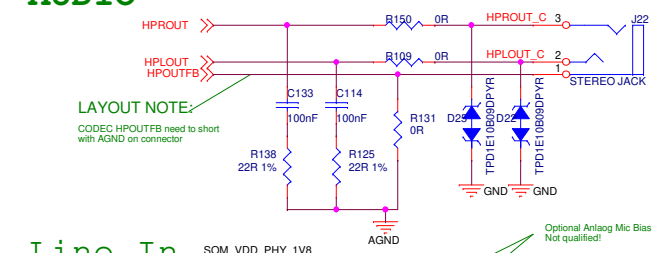


MIPI-CSI0 + MIPI-CSI1

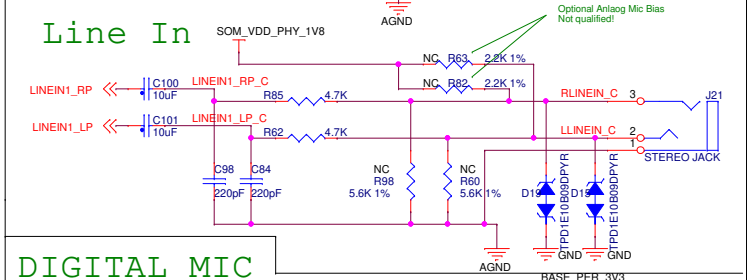


AUDIO

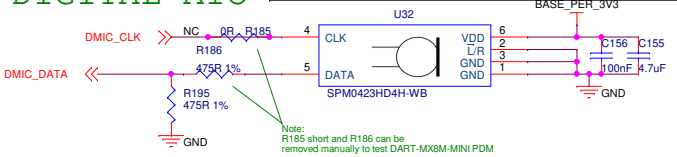
Headphones



Line In



DIGITAL MIC



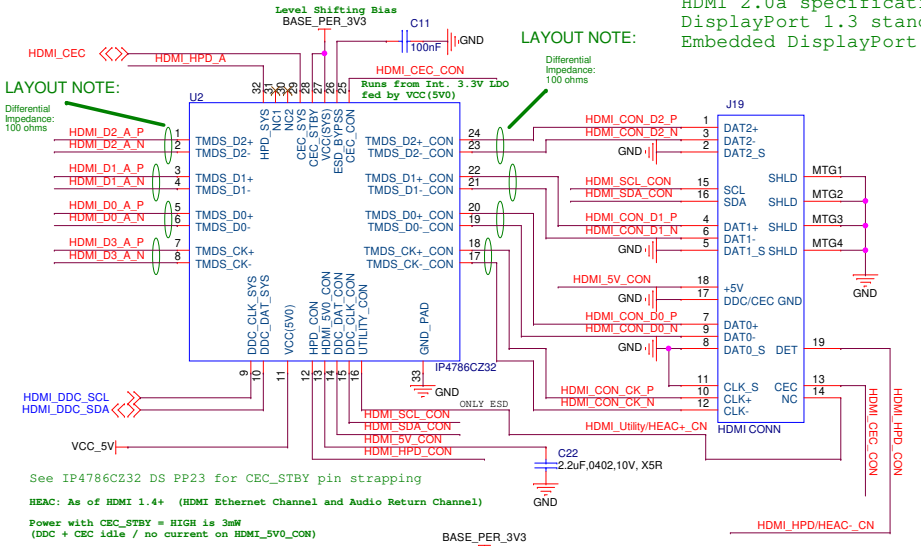
Variscite

Title: 05. ETH, uSD, AUDIO, MIPI-CSI			
Size: A3	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Monday, August 26, 2019		Sheet 7 of 15	

06. HDMI, eDP

HDMI PORT

HDMI 2.0a specification (HDMI.org)
DisplayPort 1.3 standard (VESA.org)
Embedded DisplayPort 1.4 standard



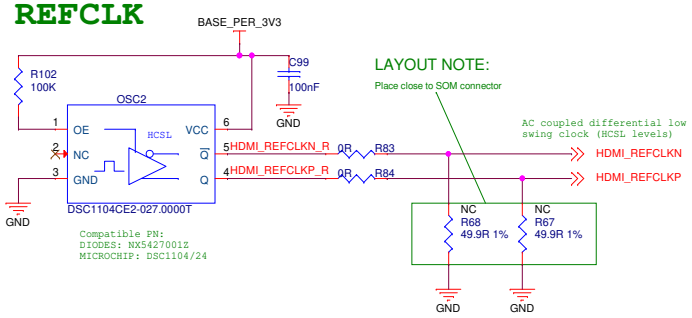
See IP4786CZ32 DS PP23 for CEC_STBY pin strapping

HEAC: As of HDMI 1.4+ (HDMI Ethernet Channel and Audio Return Channel)

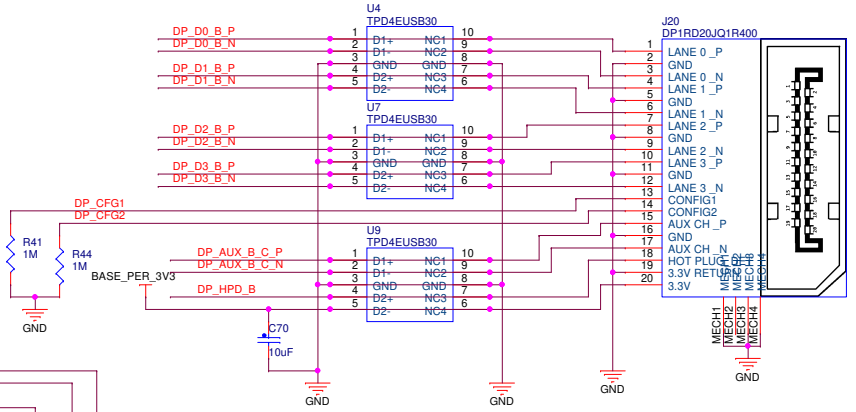
Power with CEC_STBY = HIGH is 3mW
(DDC + CEC idle / no current on HDMI_5V0_CON)

Power with CEC_STBY = LOW is 1mW

HDMI REFCLK

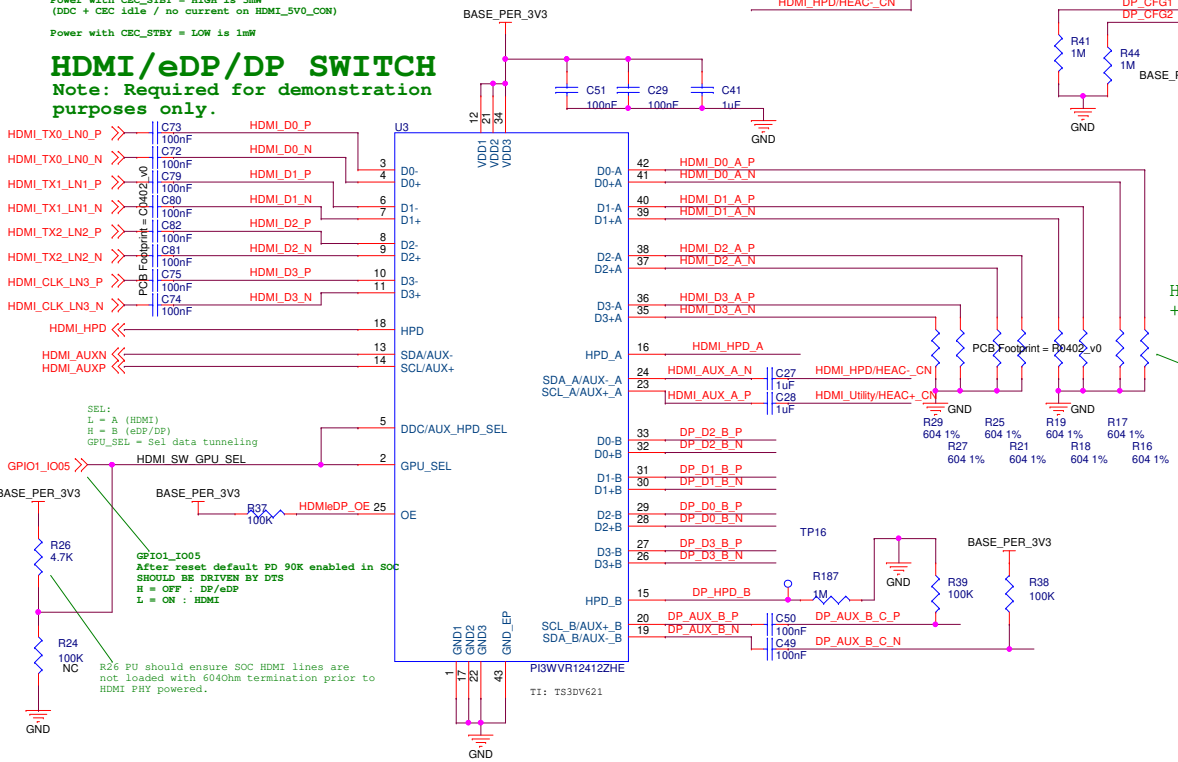


DP PORT



HDMI/eDP/DP SWITCH

Note: Required for demonstration purposes only.



HDMI PORT
+ Level Termination

Note : HDMI pull down must not be applied until VDD_PHY_1V8 is up.
Implemtation uses fact that BASE_PER_3V3 rises after all SOM power rails are up.
At boot time GPIO drives U3 switch to B state.

DP port

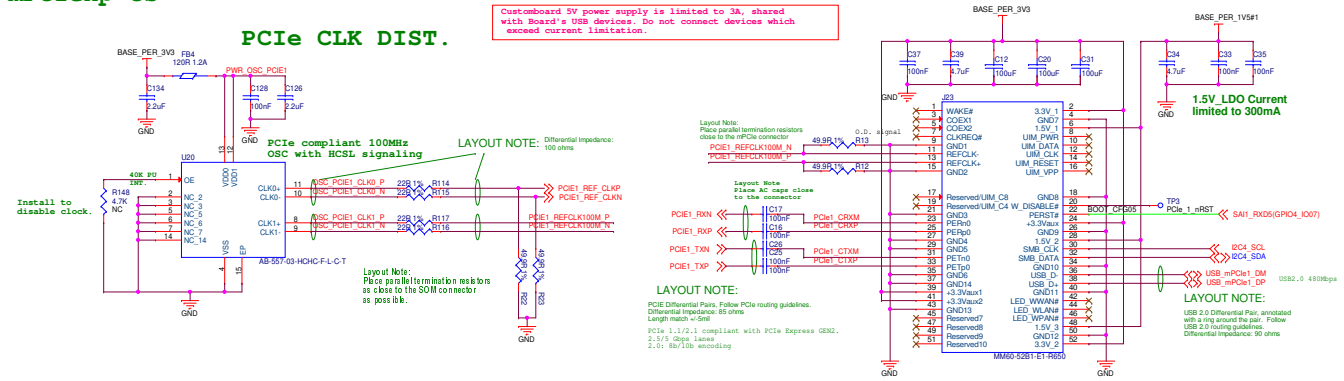
Variscite

Title: 06. HDMI , eDP			
Size: A3	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By: <Approved By>	
Date: Tuesday, August 20, 2019		Sheet 8 of 15	

07. PCIe, NAND, USB DEBUG

mPCIexp CS

PCIe CLK DIST.

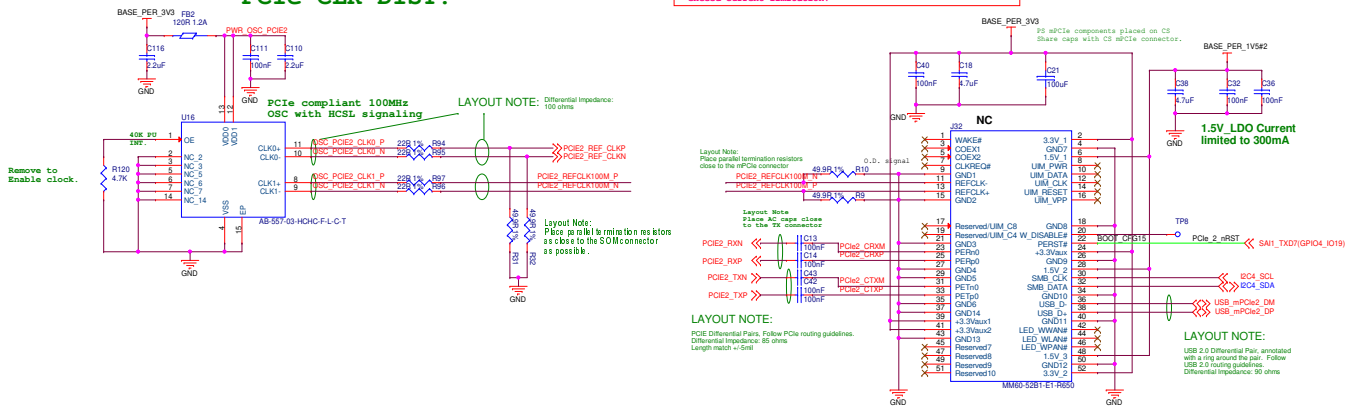


QSPI TEST POINTS ON PS

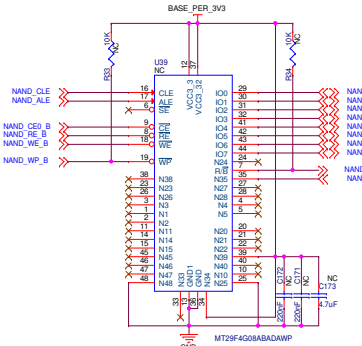


mPCIexp ON PS

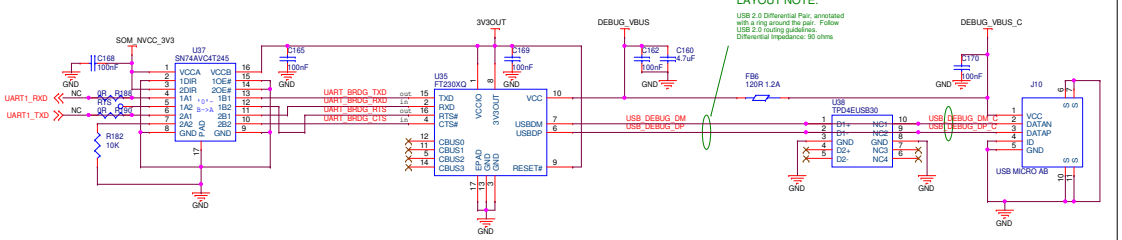
PCIe CLK DIST.



NAND



USB UART DEBUG



Title: 07. PCIe, NAND, USB DEBUG			
Size: A2	Document Number: VAR-DT8MCustomBoard	Project	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	<Approved By>
Date: Thursday, August 08, 2019		Sheet 9 of 15	

08. USB TYPE C, USB 3 HUB

USB#1

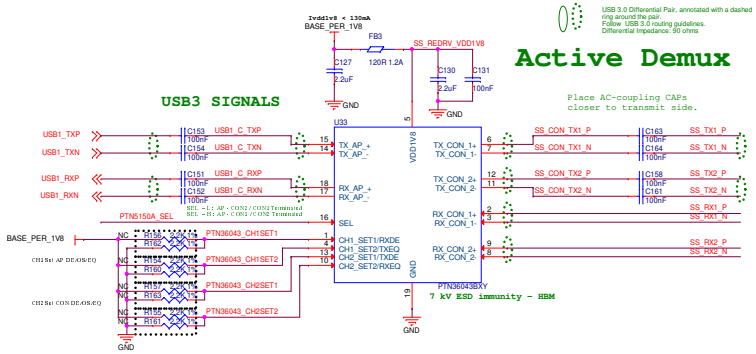
USB TYPE C OTG

LAYOUT NOTE:
USB 3.0 Differential Pair, annotated with a dashed ring around the pair.
Follow USB 3.0 routing guidelines.
Differential Impedance: 90 ohms

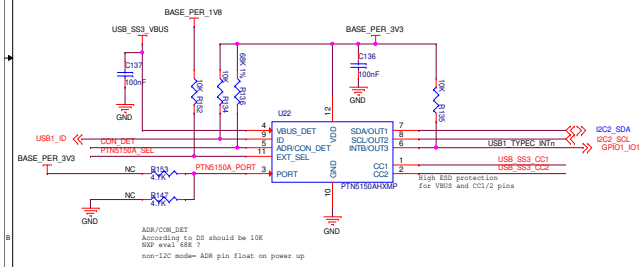
Active Demux

Place AC-coupling CAPs closer to transmit side.

USB3 SIGNALS

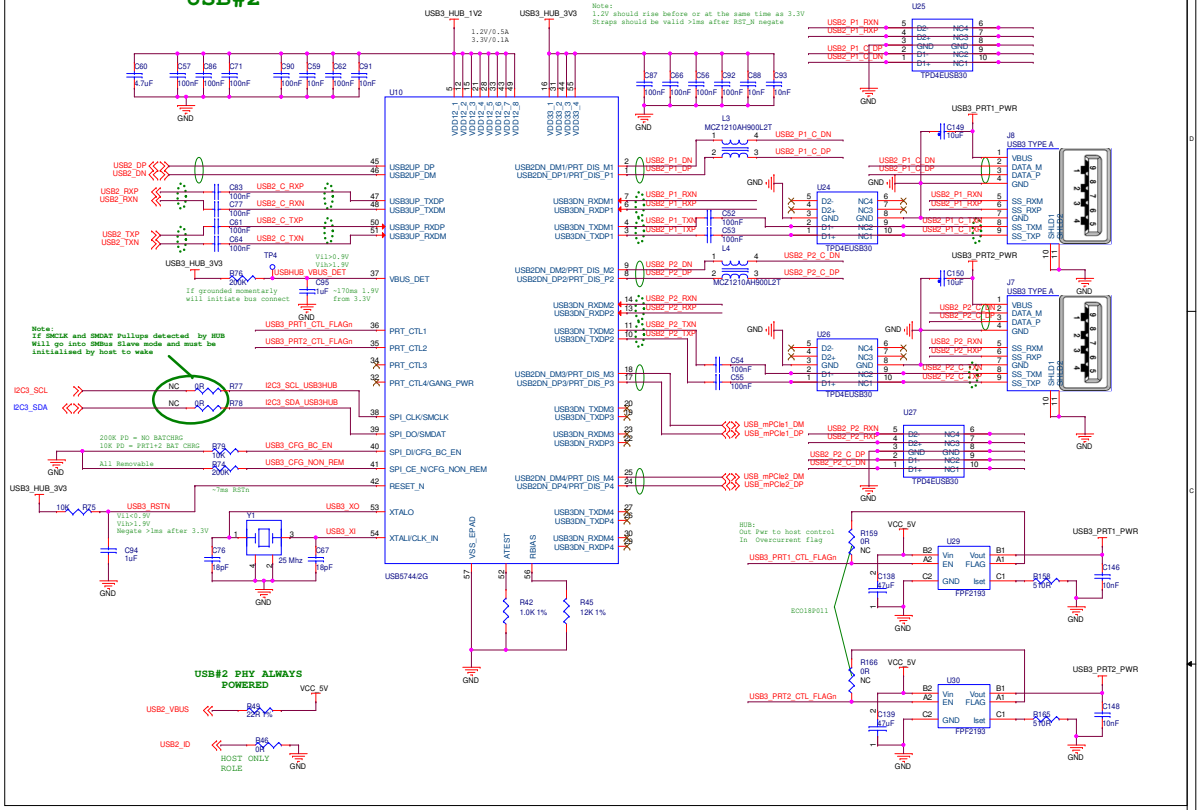


Config Channel Logic Detection & Indication of Plug Orientation

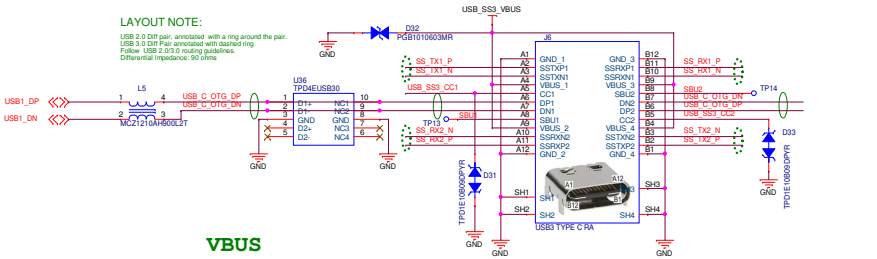


USB#2 PHY ALWAYS POWERED

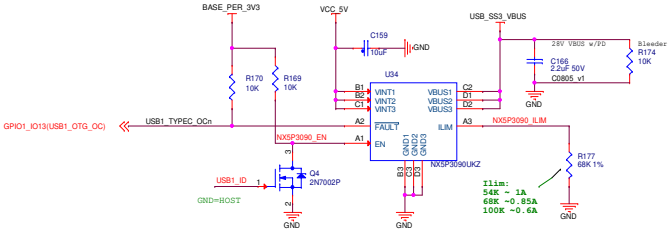
USB3.0 HUB



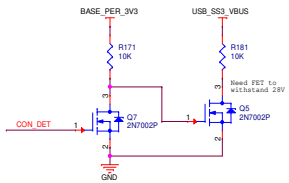
USB TYPE C OTG



5V Source Load Switch



VBUS DISCHARGE

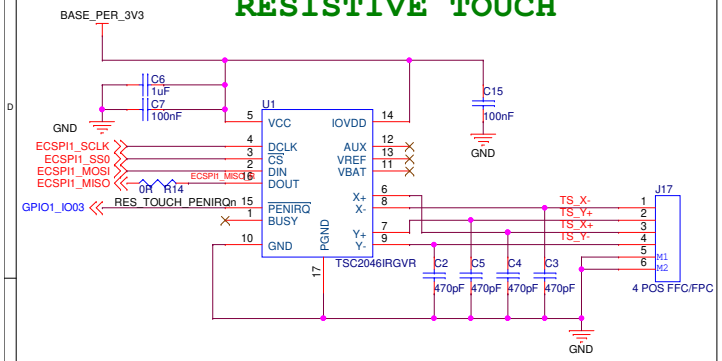


Variscite

Title: 08. USB TYPE C, USB3 HUB			
Size: A2	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R0.2
Designer: Oded A. VPC0331		Approved By:	
Date: Thursday, August 08, 2019		Sheet 10 of 15	

09. LVDS, TOUCH, JTAG, GP SW & LEDs

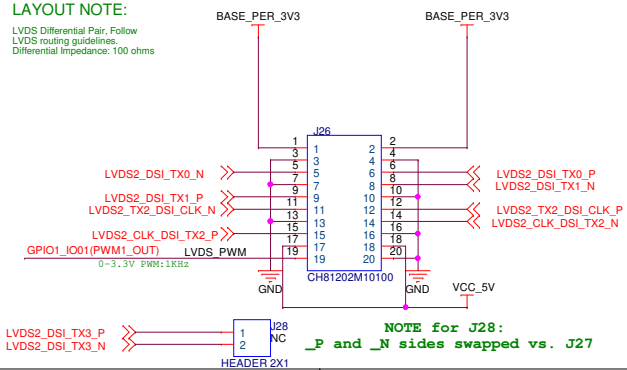
RESISTIVE TOUCH



LVDS CH1 DISPLAY

LAYOUT NOTE:

LVDS Differential Pair, Follow
LVDS routing guidelines.
Differential Impedance: 100 ohms

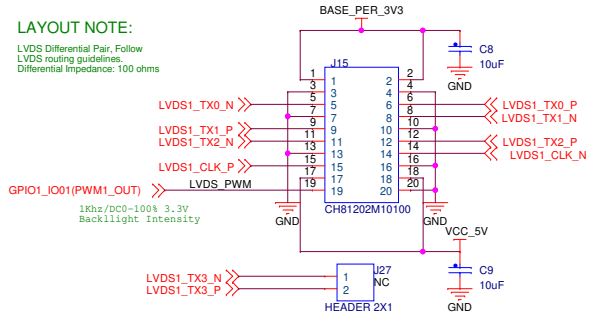


NOTE for J28:
_P and _N sides swapped vs. J27

LVDS DISPLAY CH0

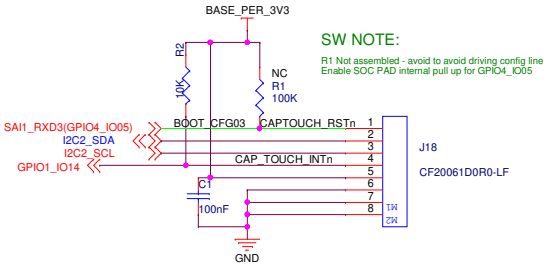
LAYOUT NOTE:

LVDS Differential Pair, Follow
LVDS routing guidelines.
Differential Impedance: 100 ohms

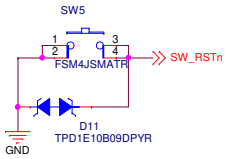


LCD: GKTW70SDAE4SE
5V/0.45A
3.3V/0.17A

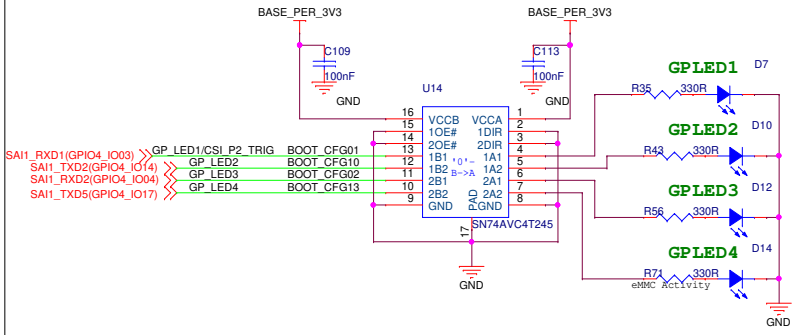
CAPACITIVE TOUCH



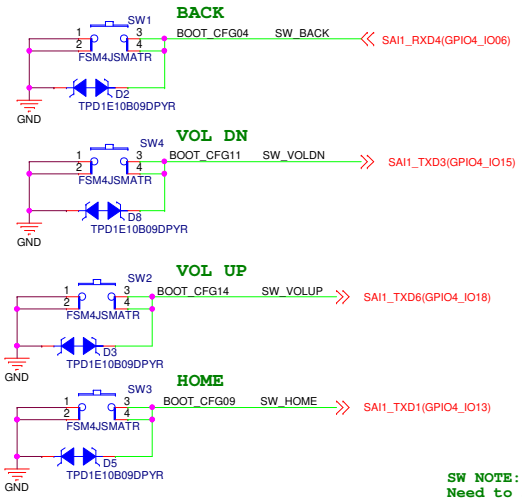
RST



GP LEDS

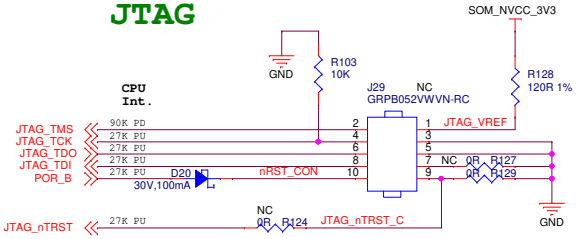


GP BUTTONS



SW NOTE:
Need to set Int. PU in SOC
on all SWitches

JTAG



The new 10 pin smaller connector is now becoming an industry standard.

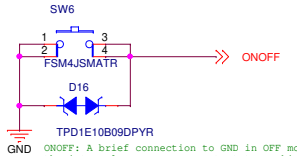
The ARM DStream comes with a 10-pin connector. If you need a ribbon, they can be purhased at Digikey:
Samtec Inc MPN: FFSD-05-D-04.00-01-N

If you need to expand to a 20-pin connector for a different JTAG device, the expander board can also be purchased at Digikey:
Olimex LTD MPN: ARM-JTAG-20-10

https://www.digikey.com/products/en/development-boards-kits-programmers/accessories/783?k=JTAG%20adapter

ON/OFF

PU Included on SOM



ONOFF: A brief connection to GND in OFF mode causes the internal power management state machine to change state to ON. In ON mode, a brief connection to GND generates an interrupt (intended to initiate a software-controllable power-down). An approximate 5 second or more connection to GND causes a forced OFF.

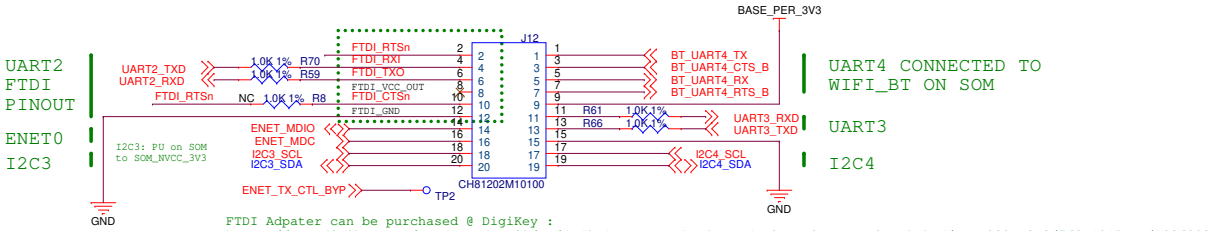
Not used leave NC

Variscite

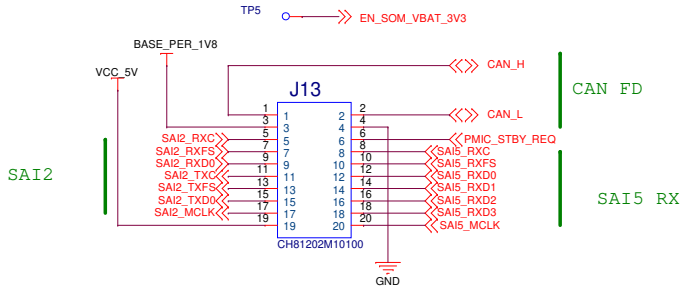
Title	09. LVDS, TOUCH, JTAG, GP SW & LEDs		
Size A3	Document Number VAR-DT8MCustomBoard	Project VAR-DT8MCustomBoard	Rev 1.4_R1.7
Designer:	Oded A. VPC0331	Approved By:	
Date:	Thursday, August 08, 2019	Sheet 11 of 15	

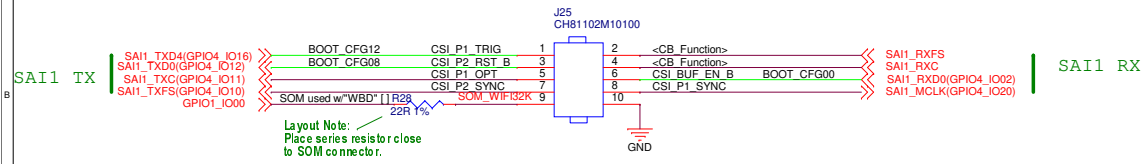
10. HEADERS, Pull Ups

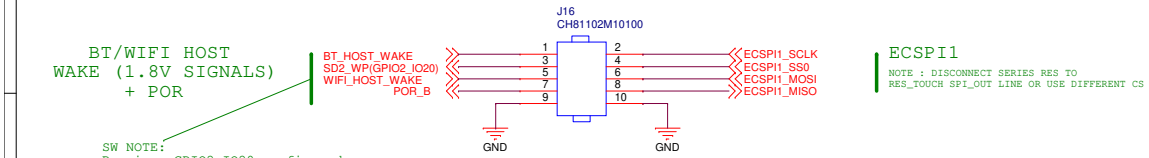
3.3V HEADERS

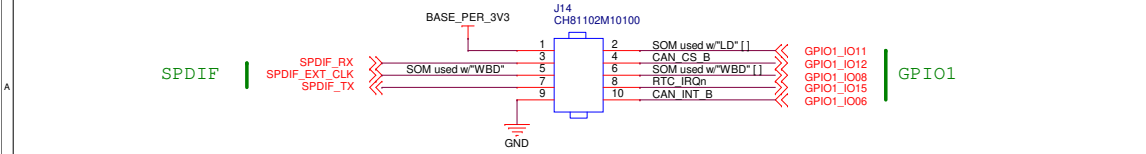


FTDI Adpater can be purchased @ DigiKey :
https://www.digikey.com/product-detail/en/ftdi-future-technology-devices-international-ltd/TTL-232R-3V3/768-1015-ND/1836393
See pinout: http://www.ftdichip.com/Support/Documents/DataSheets/Cables/DS_TTL-232R_CABLES.pdf

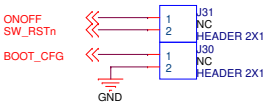






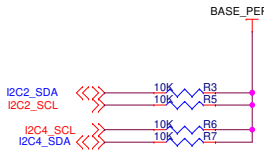


USED FOR VARISCITE TESTING



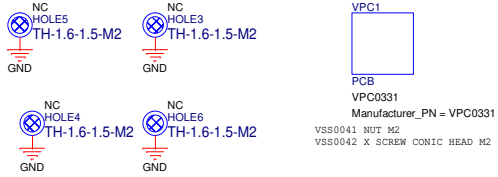
Note:
Need to set SD-eMMC Selector to OFF for remote testing

I2C PULL UP

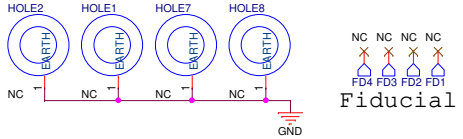


MECHANICS

SOM MOUNTING STANDOFF
Place on TOP



CHASSIS HOLES



Variscite

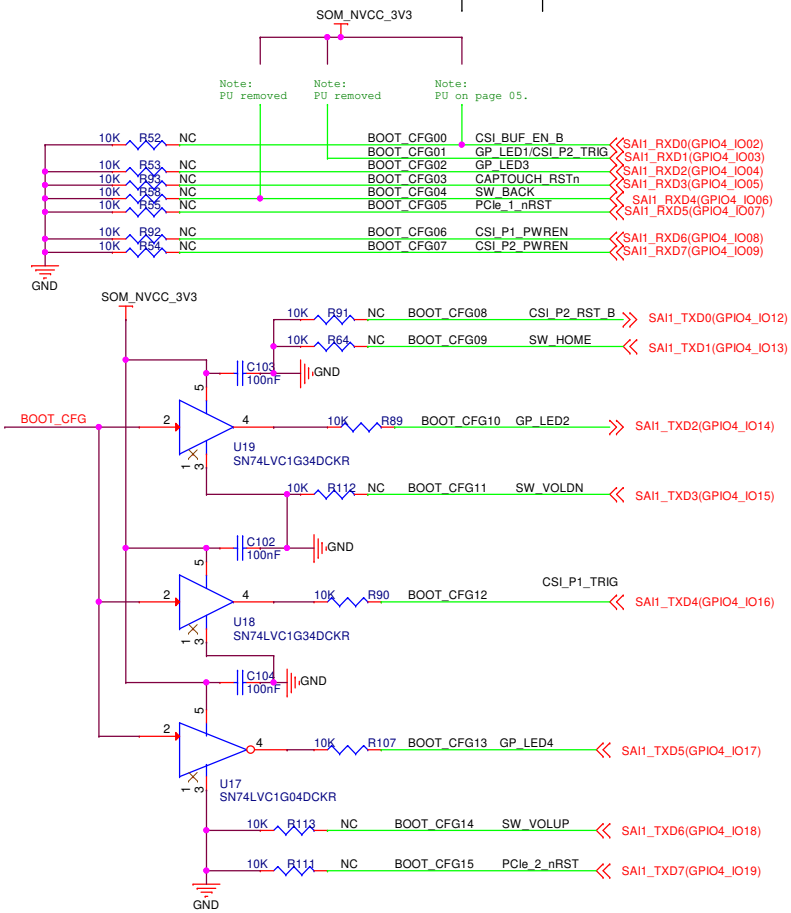
Title: 10. HEADERS, Mechanics, Pull Ups			
Size: A3	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Thursday, August 08, 2019		Sheet 12 of 15	

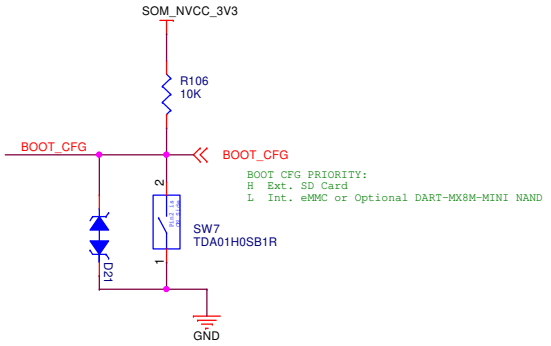
11. BOOT CONFIG & MODE

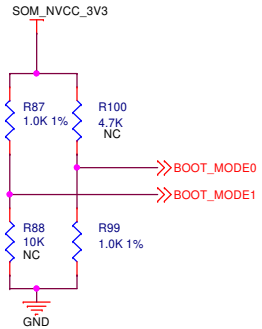
			INT. BOOT	EXT. BOOT	
SAI1_RXD0(GPIO4_IO02)	CSI_BUF_EN_B	BOOT_CFG00	0	0	Need to Enable PU in DTS; See pp. 5
SAI1_RXD1(GPIO4_IO03)	GP_LED1/CSI_P2_TRIG	BOOT_CFG01	0	0	
SAI1_RXD2(GPIO4_IO04)	GP_LED3	BOOT_CFG02	0	0	
SAI1_RXD3(GPIO4_IO05)	CAPTOUCH_RSTn	BOOT_CFG03	0	0	
SAI1_RXD4(GPIO4_IO06)	SW_BACK	BOOT_CFG04	0	0	
SAI1_RXD5(GPIO4_IO07)	PCIe_1_nRST	BOOT_CFG05	0	0	
SAI1_RXD6(GPIO4_IO08)	CSI_P1_PWREN	BOOT_CFG06	0	0	
SAI1_RXD7(GPIO4_IO09)	CSI_P2_PWREN	BOOT_CFG07	0	0	
SAI1_TXD0(GPIO4_IO12)	CSI_P2_RST_B	BOOT_CFG08	0	0	
SAI1_TXD1(GPIO4_IO13)	SW_HOME	BOOT_CFG09	0	0	
SAI1_TXD2(GPIO4_IO14)	GP_LED2	BOOT_CFG10	0	1	
SAI1_TXD3(GPIO4_IO15)	SW_VOLDN	BOOT_CFG11	0	0	
SAI1_TXD4(GPIO4_IO16)	CSI_P1_TRIG	BOOT_CFG12	0	1	
SAI1_TXD5(GPIO4_IO17)	GP_LED4	BOOT_CFG13	1	0	
SAI1_TXD6(GPIO4_IO18)	SW_VOLUP	BOOT_CFG14	0	0	
SAI1_TXD7(GPIO4_IO19)	PCIe_2_nRST	BOOT_CFG15	0	0	

Notes:
a. Sampled on rising edge of POR_B
b. 90K ohm Int. SOC PD during POR_B and after on BOOT CFG[15:0] and BOOTMODE[1:0]
c. BOOT_MODE[1:0] ="10" is Internal Boot - Always used.
d. Active boot cfg for one dip sw sel EXTERNAL/INTERNAL

DART-MX8M-MINI Notes:
a. Internal boot can be eMMC or NAND(when it is released)
b. Boot config lines do not follow the Mini datasheet in full DART-MX8M-MINI have added logic to be compatible to DART-MX8M
c. Need to modify R90 to 10K (R89 and R107 not a must)







Title: 11. BOOT CONFIG & MODE		
Size: B	Document Number: VAR-DT8MCustomBoard	Rev: 1.4_R17
Date: Thursday, August 08, 2019	Sheet 13 of 15	

12. PINMUX J1 & J2 & J3

	ALT0	ALT1	ALT2	ALT3	ALT4	ALT5	ALT6	ALT IC	CB_FUNCTION	
NAND_DATA00	NAND_DATA00	QSPIA_DATA0 [QSPIA_DATA0_1V8]				GPIO3_IO06 [GPIO3_IO06_1V8]			[SOM used w/NAND]	J1
NAND_DATA01	NAND_DATA01	QSPIA_DATA1 [QSPIA_DATA1_1V8]				GPIO3_IO07 [GPIO3_IO07_1V8]			[SOM used w/NAND]	
NAND_DATA02	NAND_DATA02	QSPIA_DATA2 [QSPIA_DATA2_1V8]				GPIO3_IO08 [GPIO3_IO08_1V8]			[SOM used w/NAND]	
NAND_DATA03	NAND_DATA03	QSPIA_DATA3 [QSPIA_DATA3_1V8]				GPIO3_IO09 [GPIO3_IO09_1V8]			[SOM used w/NAND]	
NAND_DATA04	NAND_DATA04 [NC]	QSPIB_DATA0 [ ]				GPIO3_IO10 [ ]				
NAND_DATA05	NAND_DATA05 [NC]	QSPIB_DATA1 [ ]				GPIO3_IO11 [ ]				
NAND_DATA06	NAND_DATA06 [CLKIN2_1V8]	QSPIB_DATA2 [ ]				GPIO3_IO12 [ ]				
NAND_DATA07	NAND_DATA07 [CLKIN1_1V8]	QSPIB_DATA3 [ ]				GPIO3_IO13 [ ]				
NAND_CE0_B	NAND_CE0_B	QSPIA_SS0_B [QSPIA_SS0_B_1V8]				GPIO3_IO01 [GPIO3_IO01_1V8]			[SOM used w/NAND]	
NAND_CE2_B	NAND_CE2_B [CLKOUT2_1V8]	QSPIB_SS0_B [ ]				GPIO3_IO03 [ ]				
NAND_DQS	NAND_DQS	QSPIA_DQS [QSPIA_DQS_1V8]				GPIO3_IO14 [GPIO3_IO14_1V8]			[SOM used w/NAND]	
NAND_ALE	NAND_ALE	QSPIA_SCLK [QSPIA_SCLK_1V8]				GPIO3_IO00 [GPIO3_IO00_1V8]			[SOM used w/NAND]	
NAND_WP_B	NAND_WP_B [NC]					GPIO3_IO18 [ ]				
NAND_WE_B	NAND_WE_B [NC]					GPIO3_IO17 [ ]				
NAND_RE_B	NAND_RE_B [GPIO1_IO07]	QSPIB_DQS [ENET_MDIO]				GPIO3_IO15 [ ]	[EXT_CLK4]		<CB_Function>	
NAND_CLE	NAND_CLE [CLKOUT1_1V8]	QSPIB_SCLK [ ]				GPIO3_IO05 [ ]				
NAND_READY_B	NAND_READY_B [NC]					GPIO3_IO16 [ ]				
CONN_SD2_CLK	SD2_CLK					GPIO2_IO13				
CONN_SD2_DATA0	SD2_DATA0					GPIO2_IO15				
CONN_SD2_DATA1	SD2_DATA1					GPIO2_IO16				
CONN_SD2_DATA2	SD2_DATA2					GPIO2_IO17				
CONN_SD2_DATA3	SD2_DATA3					GPIO2_IO18				
CONN_SD2_CMD	SD2_CMD					GPIO2_IO14				
CONN_SD2_CD_B	SD2_CD_B					GPIO2_IO12				
CONN_SD2_nRST	SD2_RESET_B					GPIO2_IO19			SD2_PWR_CTRL	
SD2_WP(GPIO2_IO20)	SD2_WP [SD1_DATA7]					GPIO2_IO20 [GPIO2_IO09_1V8]				
ETH_TRX0_N	ENET_TD2	ENET_TX_CLK_IN/ENET_REF_CLK_ROOT_OUT				GPIO1_IO19		ETH_TRX0_N		
ETH_TRX0_P	ENET_TD3					GPIO1_IO18		ETH_TRX0_P		
ETH_TRX1_N	ENET_TD0					GPIO1_IO21		ETH_TRX1_N		
ETH_TRX1_P	ENET_TD1					GPIO1_IO20		ETH_TRX1_P		
ETH_TRX2_N	ENET_RD1					GPIO1_IO27		ETH_TRX2_N		
ETH_TRX2_P	ENET_RD0					GPIO1_IO26		ETH_TRX2_P		
ETH_TRX3_N	ENET_RD3					GPIO1_IO29		ETH_TRX3_N		
ETH_TRX3_P	ENET_RD2					GPIO1_IO28		ETH_TRX3_P		
ENET_TX_CTL_8YP	ENET_TX_CTL					GPIO1_IO22		NC		
LED_LINK10_100	ENET_TXC	ENET_TX_ER				GPIO1_IO23		LED_LINK10_100		
LED_LINK1000	ENET_RXC	ENET_RX_ER				GPIO1_IO25		LED_LINK1000		
LED_ACT	ENET_RX_CTL					GPIO1_IO24		LED_ACT		
ENET_MDIO	ENET_MDIO					GPIO1_IO17			SOM used w/"EC"	
ENET_MDC	ENET_MDC					GPIO1_IO16			SOM used w/"EC"	
GPIO1_IO00	GPIO1_IO00	ENET_PHY_REF_CLK_ROOT_OUT				REF_CLK_32K	EXT_CLK1		SOM used w/"WBD" [ ]	
I2C4_SCL	I2C4_SCL	PWM2_OUT	PCIE1_CLKREQ_B			GPIO5_IO20				
I2C4_SDA	I2C4_SDA	PWM1_OUT	PCIE2_CLKREQ_B [ ]			GPIO5_IO21				
HPLOUT	SAI3_RXD	GPT1_COMPARE1	SAI5_RXD0			GPIO4_IO30		HPLOUT		J2
HPROUT	SAI3_TXC	GPT1_COMPARE2	SAI5_RXD2		[UART2_TXD]	GPIO5_IO00		HPROUT		
HPOUTFB	SAI3_RXFS	GPT1_CAPTURE1	SAI5_RXFS	[SAI3_RXD1]		GPIO4_IO28		HPOUTFB		
LINEIN1_LP	SAI3_RXC	GPT1_CAPTURE2 [GPT1_CLK]	SAI5_RXC		[UART2_CTS_B]	GPIO4_IO29		LINEIN1_LP		
LINEIN1_RP	SAI3_TXFS	GPT1_CLK [GPT1_CAPTURE2]	SAI5_RXD1	[SAI3_TXD1]	[UART2_RXD]	GPIO4_IO31		LINEIN1_RP		
DMIC_CLK	SAI3_TXD	GPT1_COMPARE3	SAI5_RXD3			GPIO5_IO01		DMIC_CLK		
DMIC_DATA	SAI3_MCLK	PWM4_OUT	SAI5_MCLK			GPIO5_IO02		DMIC_DATA		
BT_UART4_TX	ECSPI2_MOSI	UART4_TXD				GPIO5_IO11			SOM used w/"WBD"	
BT_UART4_CTS_B	ECSPI2_MISO	UART4_CTS_B				GPIO5_IO12			SOM used w/"WBD"	
BT_UART4_RX	ECSPI2_SCLK	UART4_RXD				GPIO5_IO10			SOM used w/"WBD"	
BT_UART4_RTS_B	ECSPI2_SS0	UART4_RTS_B				GPIO5_IO13			SOM used w/"WBD"	
GPIO1_IO02(nWDOG)	GPIO1_IO02	WDOG_B				[WDOG_ANY]			WDOGin	
I2C2_SDA	I2C2_SDA	ENET1_1588_EVENT1_OUT				GPIO5_IO17				
I2C2_SCL	I2C2_SCL	ENET1_1588_EVENT1_IN				GPIO5_IO16				
ECSPI1_SCLK	ECSPI1_SCLK	UART3_RXD				GPIO5_IO06				
ECSPI1_SS0	ECSPI1_SS0	UART3_RTS_B				GPIO5_IO09				
ECSPI1_MISO	ECSPI1_MISO	UART3_CTS_B				GPIO5_IO08				
ECSPI1_MOSI	ECSPI1_MOSI	UART3_TXD				GPIO5_IO07				
UART1_RXD	UART1_RXD	ECSPI3_SCLK				GPIO5_IO22				
UART1_TXD	UART1_TXD	ECSPI3_MOSI				GPIO5_IO23				
UART2_RXD	UART2_RXD	ECSPI3_MISO				GPIO5_IO24				
UART2_TXD	UART2_TXD	ECSPI3_SS0				GPIO5_IO25				
UART3_RXD	UART3_RXD	UART1_CTS_B				GPIO5_IO26				
UART3_TXD	UART3_TXD	UART1_RTS_B				GPIO5_IO27				
SAI2_MCLK	SAI2_MCLK	SAI5_MCLK				GPIO4_IO27				
SAI2_RXFS	SAI2_RXFS	SAI5_TXFS	[SAI5_TXD1]	[SAI2_RXD1]	[UART1_TXD]	GPIO4_IO21				
SAI2_RXC	SAI2_RXC	SAI5_TXC			[UART1_RXD]	GPIO4_IO22				
SAI2_TXFS	SAI2_TXFS	SAI5_TXD1		[SAI2_TXD1]	[UART1_CTS_B]	GPIO4_IO24				
SAI2_TXC	SAI2_TXC	SAI5_TXD2				GPIO4_IO25				
SAI2_RXD0	SAI2_RXD0	SAI5_TXD0			[UART1_RTS_B]	GPIO4_IO23				
SAI2_TXD0	SAI2_TXD0	SAI5_TXD3				GPIO4_IO26				
SAI5_MCLK	SAI5_MCLK	SAI1_TXC	SAI4_MCLK [ ]			GPIO3_IO25				
SAI5_RXFS	SAI5_RXFS	SAI1_TXD0				GPIO3_IO19				
SAI5_RXC	SAI5_RXC	SAI1_TXD1			[PDM_CLK]	GPIO3_IO20				
SAI5_RXD0	SAI5_RXD0	SAI1_TXD2			[PDM_BIT0]	GPIO3_IO21				
SAI5_RXD1	SAI5_RXD1	SAI1_TXD3	SAI1_TXFS	SAI5_TXFS	[PDM_BIT1]	GPIO3_IO22				
SAI5_RXD2	SAI5_RXD2	SAI1_TXD4	SAI1_TXFS	SAI5_TXC	[PDM_BIT2]	GPIO3_IO23				
SAI5_RXD3	SAI5_RXD3	SAI1_TXD5	SAI1_TXFS	SAI5_TXD0	[PDM_BIT3]	GPIO3_IO24				
SAI1_MCLK(GPIO4_IO20)	SAI1_MCLK	SAI5_MCLK	SAI1_TXC	[PDM_CLK]		GPIO4_IO20			CSI_P1_SYNC	
SAI1_RXFS	SAI1_RXFS	SAI5_RXFS				GPIO4_IO00				
SAI1_RXC	SAI1_RXC	SAI5_RXC				GPIO4_IO01				
SAI1_TXFS(GPIO4_IO10)	SAI1_TXFS	SAI5_TXFS				GPIO4_IO10			CSI_P2_SYNC	
SAI1_TXC(GPIO4_IO11)	SAI1_TXC	SAI5_TXC				GPIO4_IO11			CSI_P1_OPT	
SAI1_RXD0(GPIO4_IO02)	SAI1_RXD0	SAI5_RXD0	[SAI1_TXD1]	[PDM_BIT0]		GPIO4_IO02	BOOT_CFG00		CSI_BUF_EN_B	
SAI1_RXD1(GPIO4_IO03)	SAI1_RXD1	SAI5_RXD1		[PDM_BIT1]		GPIO4_IO03	BOOT_CFG01		GP_LED1/CSI_P2_TRIG	
SAI1_RXD2(GPIO4_IO04)	SAI1_RXD2	SAI5_RXD2		[PDM_BIT2]		GPIO4_IO04	BOOT_CFG02		GP_LED3	
SAI1_RXD3(GPIO4_IO05)	SAI1_RXD3	SAI5_RXD3		[PDM_BIT3]		GPIO4_IO05	BOOT_CFG03		CAPTOUCH_RSTn	
SAI1_RXD4(GPIO4_IO06)	SAI1_RXD4	SAI6_TXC	SAI6_RXC			GPIO4_IO06	BOOT_CFG04		SW_BACK	
SAI1_RXD5(GPIO4_IO07)	SAI1_RXD5	SAI6_TXD0	SAI6_RXD0	SAI1_RXFS		GPIO4_IO07	BOOT_CFG05		PCIe_1_nRST	
SAI1_RXD6(GPIO4_IO08)	SAI1_RXD6	SAI6_TX_SYNC	SAI6_RXFS			GPIO4_IO08	BOOT_CFG06		CSI_P1_PWREN	
SAI1_RXD7(GPIO4_IO09)	SAI1_RXD7	SAI6_MCLK	SAI1_TXFS	SAI1_TXD4		GPIO4_IO09	BOOT_CFG07		CSI_P2_PWREN	
SAI1_TXD0(GPIO4_IO12)	SAI1_TXD0	SAI5_TXD0				GPIO4_IO12	BOOT_CFG08		CSI_P2_RST_B	
SAI1_TXD1(GPIO4_IO13)	SAI1_TXD1	SAI5_TXD1				GPIO4_IO13	BOOT_CFG09		SW_HOME	
SAI1_TXD2(GPIO4_IO14)	SAI1_TXD2	SAI5_TXD2				GPIO4_IO14	BOOT_CFG10		GP_LED2	
SAI1_TXD3(GPIO4_IO15)	SAI1_TXD3	SAI5_TXD3				GPIO4_IO15	BOOT_CFG11		SW_VOLDN	
SAI1_TXD4(GPIO4_IO16)	SAI1_TXD4	SAI6_RXC	SAI6_TXC			GPIO4_IO16	BOOT_CFG12		CSI_P1_TRIG	
SAI1_TXD5(GPIO4_IO17)	SAI1_TXD5	SAI6_RXD0	SAI6_TXD0			GPIO4_IO17	BOOT_CFG13		GP_LED4	
SAI1_TXD6(GPIO4_IO18)	SAI1_TXD6	SAI6_RXFS	SAI6_TXFS			GPIO4_IO18	BOOT_CFG14		SW_VOLUP	
SAI1_TXD7(GPIO4_IO19)	SAI1_TXD7	SAI6_MCLK		[PDM_CLK]		GPIO4_IO19	BOOT_CFG15		PCIe_2_nRST	
UART4_TXD(GPIO5_IO29)	UART4_TXD	UART2_RTS_B	PCIE2_CLKREQ_B [ ]			GPIO5_IO29			CSI_P2_OPT	J3
UART4_RXD(GPIO5_IO28)	UART4_RXD	UART2_CTS_B	PCIE1_CLKREQ_B			GPIO5_IO28			CSI_P1_RST_B	
SPDIF_RX	SPDIF_RX	PWM2_OUT				GPIO5_IO04				
SPDIF_TX	SPDIF_TX	PWM3_OUT				GPIO5_IO03				
SPDIF_EXT_CLK	SPDIF_EXT_CLK	PWM1_OUT				GPIO5_IO05			SOM used w/"WBD"	
I2C3_SDA	I2C3_SDA	PWM3_OUT	GPT3_CLK			GPIO5_IO19			SOM Shared	
I2C3_SCL	I2C3_SCL	PWM4_OUT	GPT2_CLK			GPIO5_IO18			SOM Shared	
GPIO1_IO01(PWM1_OUT)	GPIO1_IO01	PWM1_OUT				REF_CLK_24M	EXT_CLK2		LVDS_PWM	
GPIO1_IO03	GPIO1_IO03	USDHC1_VSELECT					XTAL_OK [ ]		RES_TOUCH_PENIRQn	
GPIO1_IO05	GPIO1_IO05	M4_NMI				PMIC_READY			HDMI_SW_GPU_SEL	
GPIO1_IO06	GPIO1_IO06	ENET1_MDC				SD1_CD_B	EXT_CLK3		CAN_INT_B	
GPIO1_IO08	GPIO1_IO08	ENET1_1588_EVENT0_IN				SD2_RESET_B			SOM used w/"WBD" [ ]	
GPIO1_IO10	GPIO1_IO10	USB1_OTG_ID							USB1_TYPEC_INTn	
GPIO1_IO11	GPIO1_IO11	USB2_OTG_ID				PMIC_READY [ ]			SOM used w/"LD" [ ]	
GPIO1_IO12	GPIO1_IO12	USB1_OTG_PWR							CAN_CS_B	
GPIO1_IO13(USB1_OTG_OC)	GPIO1_IO13	USB1_OTG_OC				PWM2_OUT			USB1_TYPEC_OCn	
GPIO1_IO14	GPIO1_IO14	USB2_OTG_PWR				PWM3_OUT	[CLKO1]		CAP_TOUCH_INTn	
GPIO1_IO15	GPIO1_IO15	USB2_OTG_OC				PWM4_OUT	CLKO2 [CLKO1]		RTC_IRQn	
LVDS2_CLK_DSI_TX2_P	DSI_TX2_P							LVDS2_CLK_P		
LVDS2_CLK_DSI_TX2_N	DSI_TX2_N							LVDS2_CLK_N		
LVDS2_DSI_TX0_P	DSI_TX0_P							LVDS2_TX0_P		
LVDS2_DSI_TX0_N	DSI_TX0_N							LVDS2_TX0_N		
LVDS2_DSI_TX1_P	DSI_TX1_P							LVDS2_TX1_P		
LVDS2_DSI_TX1_N	DSI_TX1_N							LVDS2_TX1_N		
LVDS2_TX2_DSI_CLK_P	DSI_CLK_P							LVDS2_TX2_P		
LVDS2_TX2_DSI_CLK_N	DSI_CLK_N							LVDS2_TX2_N		
LVDS2_DSI_TX3_P	DSI_TX3_P							LVDS2_TX3_P		
LVDS2_DSI_TX3_N	DSI_TX3_N							LVDS2_TX3_N		
LVDS1_CLK_P	NC							LVDS1_CLK_P		
LVDS1_CLK_N	NC							LVDS1_CLK_N		
LVDS1_TX0_P	NC							LVDS1_TX0_P		
LVDS1_TX0_N	NC							LVDS1_TX0_N		
LVDS1_TX1_P	NC							LVDS1_TX1_P		
LVDS1_TX1_N	NC							LVDS1_TX1_N		
LVDS1_TX2_P	NC							LVDS1_TX2_P		
LVDS1_TX2_N	NC							LVDS1_TX2_N		
LVDS1_TX3_P	NC							LVDS1_TX3_P		
LVDS1_TX3_N	NC							LVDS1_TX3_N		

NOTE:

1. NOTE CB_FUNCTION FOR "SOM used", WHICH MEANS SIGNAL USED ON SOME SOMs WITH OPTIONAL OREDERABLE CONFIGURATIONS. PLEASE REFER TO SOM DATASHEET FOR FURTHER DETAILS.
2. ALT0 TO ALT6 DENOTE ALTERNATE FUNCTIONS OF SOC PINS WHEN USED.
3. ALT IC DENOTE AN ALTERNATE FUNCTION WHEN RELEVANT SOM ORDERABLE CONFIGURATION CHOOSEN. FUNCTION GENERATED BY ALTERNATIVE IC WHEN USED. WHEN DESIGN WITH ALT IC - ALT0 TO ALT6 CANNOT BE USED!
4. GREEN NETS DENOTE PINS USED FOR BOOT CONFIGURATION DURING POWER UP. CARE SHOULD BE GIVEN NOT TO DRIVE THESE LINES BEFORE RISE OF POR_B+1ms.

DART-MX8M-Mini differences:

[ ] square brackets are related alternate function to MINI only.
Empty brackets means function removed.

1. PMIC_ON_REQ + PMIC_STBY_REQ + POR_B + ONOFF run @ 1.8V
2. PCIe port 2 does not exist
3. MIPI-CSI port 2 does not exist
4. USB3.0 does not exist on port 1 and 2
5. HDMI does not exist
6. QSPIB does not exist
7. QSPIA exist only with eMMC on SOM and run @ 1.8V

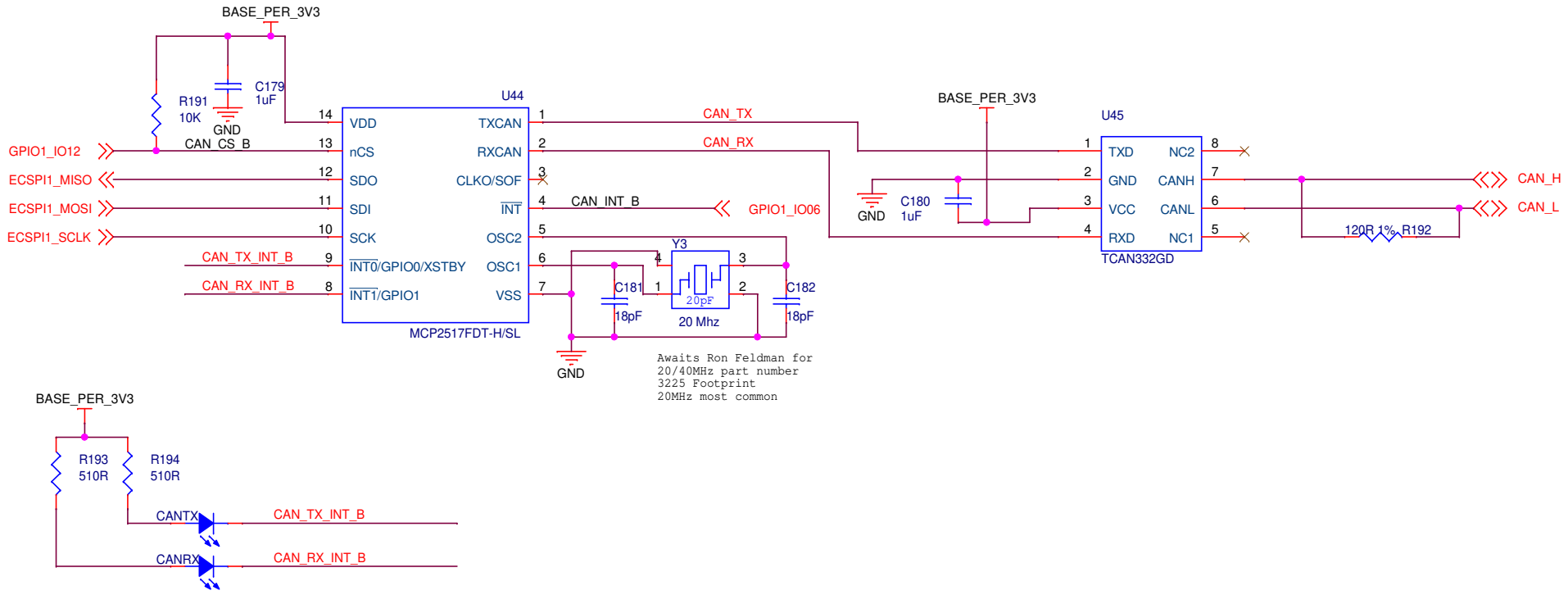
Variscite

Title: 12. PINMUX J1 & J2 & J3

Size	Document Number	Project	Rev
A2	VAR-DT8MCustomBoard	VAR-DT8MCustomBoard	1.4_R1.7

Designer: Oded A. VPC0331 | Approved By:
Date: Thursday, August 08, 2019 | Sheet 14 of 15

13. CAN FD Interface



Title: CAN FD Interface			
Size: A4	Document Number: VAR-DT8MCustomBoard	Project: VAR-DT8MCustomBoard	Rev: 1.4_R1.7
Designer: Oded A. VPC0331		Approved By:	
Date: Thursday, August 08, 2019		Sheet 15 of 15	